Chapter 11
Properties of Options and Arbitrage Bounds

This chapter contains no model. Nothing in it assumes a distribution for the stock price, a volatility, or a stochastic process, and none of its results will be revised when Chapter 15 supplies a formula. Everything here follows from the absence of arbitrage, and the results are therefore stronger than anything a model can deliver: a model can be wrong, whereas these relationships hold for any model that is not internally inconsistent.

That is worth stating plainly because the temptation is to skip ahead. A reader who knows the Black–Scholes–Merton formula but not put–call parity will price options adequately and misunderstand the market: they will read a price relationship as sentiment, exercise an American call that should have been sold, or construct a "riskless" arbitrage that carries an assignment risk they did not know existed.

Chapter 10 deferred four questions to this chapter: why the option chain in Example 10.3 had to be internally consistent, why a put and a written call together behave like short stock, why dividends make early exercise of an American call rational, and why an American put is worth more than a European one. All four are answered by arbitrage arguments alone.

Section 11.1 lists what determines an option price. Sections 11.2 and 11.3 derive the bounds on a single option and across strikes; Sections 11.4 to 11.7 are put–call parity and what follows from it: synthetic positions, the implied forward, and the box spread as a financing trade. Sections 11.8 to 11.11 are early exercise: why an American call on a stock paying no dividend is never exercised early, why a put may be, what dividends change, and the parity bounds that survive for American options. Section 11.12 asks what an apparent violation actually means, and Section 11.13 who trades these relationships.

11.1 WHAT DETERMINES AN OPTION PRICE

Six quantities determine the value of a stock option: the current stock price S_0, the strike K, the time to expiration T, the volatility σ, the risk-free rate r, and the dividends expected during the option's life. Table 11.1 gives the direction of each effect; the reasoning matters more.

Table 11.1: Effect of increasing one variable, holding the others fixed.

Variable increases	European call	European put	American call	American put
Stock price S_0	+	−	+	−
Strike price K	−	+	−	+
Time to expiration T	?	?	+	+
Volatility σ	+	+	+	+
Risk-free rate r	+	−	+	−
Dividends	−	+	−	+

Figure 11.1 shows the shapes. Two are worth noting beyond their direction, though their

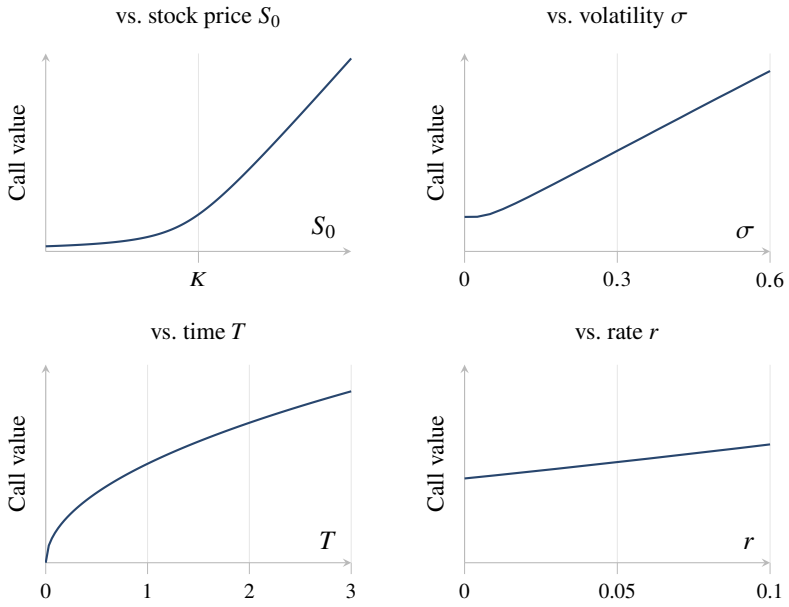


Figure 11.1: A European call, varying one variable at a time. Schematic.

shapes come from the model of Chapter 15 rather than from arbitrage. An at-the-money call's value is flat in volatility at first and then close to linear (the slope is what Chapter 18 names vega, and Chapter 19 relies on the near-linearity when it treats implied volatility as a quoted price). The dependence on time grows roughly with $\sqrt{T}$, so doubling the maturity does not double the time value.

Stock and strike prices. A call pays $\max(S_T - K, 0)$, which rises with S_T and falls with K.

Volatility. This entry separates options from every instrument in Part I and deserves care. For a forward contract, volatility is irrelevant to the price: the payoff is linear, so increased dispersion raises the good outcomes and lowers the bad ones symmetrically, and they cancel. An option's payoff is not linear. Its downside is truncated at zero, so greater dispersion raises the expected payoff without a matching penalty. *Both* calls and puts increase in value with volatility, which surprises readers who think of a call and a put as opposite bets.

Risk-free rate. Two effects run in the same direction for a call. A higher rate reduces the present value of the strike the holder will eventually pay, and it raises the forward price, $S_0 e^{rT}$ for a stock paying no dividend, on which the call is effectively written. Both increase call values and decrease put values. This is one of the few places where intuition from other markets misleads: rising rates *raise* call prices, holding the stock price fixed.

Dividends. A dividend reduces the stock price on the ex-dividend date without compensating an option holder, who receives no dividends. Calls fall in value, puts rise. Section 11.10 shows this is also what makes early exercise of American calls rational.

Time to expiration, and why two entries are question marks. An American option with more time cannot be worth less, because its holder can do everything the shorter-dated holder can do and more: the longer option's exercise opportunities are a superset.

European options do not have this property, and the reason is instructive. Consider two European calls on a stock paying a large dividend in one month, one expiring in two weeks and one in six weeks. The shorter option expires before the stock drops on the ex-dividend date; the longer one does not. The shorter option can be worth more. For a different reason,

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a European put can lose value with more time when rates are high and the put is deep in the money: the extra time defers the receipt of K. Superset reasoning requires the freedom to act, and European options do not have it.

11.2 UPPER AND LOWER BOUNDS

Before any question of what an option is worth, there is the question of what it cannot be worth.

Notation. Throughout this chapter c and p are the prices of a European call and put, and C and P the prices of the American call and put on the same stock with the same strike K and expiration T; S_0 is the current stock price and r the risk-free rate. A capital P standing alone is always the American put; the bond price of later chapters carries its two arguments, $P(t, T)$.

The assumption behind everything that follows. The chapter assumes no arbitrage, as defined at the opening of Chapter 5, and frictionless trading (borrowing and lending at r, shorting without cost, no bid–ask spread), which Section 11.12 relaxes. Each bound below is proved the same way: suppose it fails, write down the trade that then costs nothing and cannot lose, and conclude the prices cannot stand.

Upper bounds, and the exercise floor. A call gives the right to buy the stock, so it can never be worth more than the stock: $c \le S_0$ and $C \le S_0$. If it were, selling the call and buying the stock would collect $c - S_0 > 0$ today, and the share covers any exercise. A put gives the right to sell at K, so it can never be worth more than K; and since a European put pays at most K at time T, its value is capped by the present value: $p \le Ke^{-rT}$, while $P \le K$. An American option can be exercised at once, so it is never worth less than its intrinsic value: $C \ge \max(S_0 - K, 0)$ and $P \ge \max(K - S_0, 0)$.

Lower bound for a European call. Four portfolios, defined once here, carry the chapter's main proofs:

Portfolio A	one European call, plus cash of Ke^{-rT}
Portfolio B	one share
Portfolio C	one European put, plus one share
Portfolio D	cash of Ke^{-rT}

Compare A with B. The cash in A grows to exactly K by time T. If $S_T > K$ the call is exercised, using the K to buy the share, and A is worth S_T. If $S_T \le K$ the call lapses and A is worth K. So A is worth $\max(S_T, K)$, which is never less than B's S_T. A portfolio that is never worth less at T must not cost less today, so $c + Ke^{-rT} \ge S_0$, and since $c \ge 0$,

$$c \ge \max\left(S_0 - Ke^{-rT},\ 0\right) \tag{11.1}$$

Comparing Portfolio C, worth $\max(S_T, K)$, with Portfolio D, worth K, gives

$$p \ge \max\left(Ke^{-rT} - S_0,\ 0\right) \tag{11.2}$$

If dividends with present value D are paid during the option's life, the share collects them and the option does not, so S_0 becomes $S_0 - D$ in both bounds (Table 11.2). Figure 11.2 draws the region a call's bounds permit without dividends, and Example 11.1 trades a call priced below it.

Example 11.1 Arbitrage when the lower bound fails

A non-dividend-paying stock trades at \$20. A one-year European call struck at \$18 trades at \$3.00. The rate is 10%.

The lower bound (11.1) is

$$20 - 18e^{-0.10} = 20 - 16.287 = \$3.713$$

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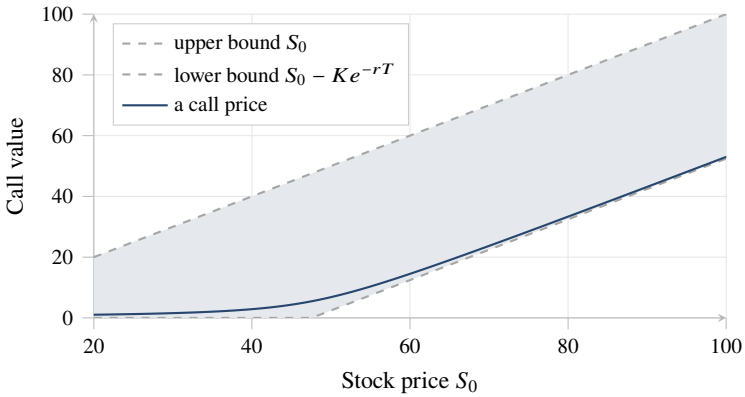


Figure 11.2: Permitted region. $K = 50$, $r = 5\%$, $T = 1$.

so \$3.00 is too cheap. Buy the call and short the stock:

$$\text{cash received} = 20 - 3 = \$17, \qquad \text{grown for one year} = 17e^{0.10} = \$18.788$$

At expiration there are two cases. If $S_T > 18$, exercise the call, pay \$18 for the share, return it, and keep $18.788 - 18 = \$0.788$. If $S_T \le 18$, let the call lapse, buy the share in the market for $S_T \le 18$, return it, and keep at least \$0.788.

The profit is at least \$0.788 in every state, from an initial investment of nothing. The mispricing is $3.713 - 3.00 = \$0.713$, and $0.713 \times e^{0.10} = \$0.788$. The guaranteed profit is exactly the mispricing carried forward at the risk-free rate, a useful check on any such calculation.

11.3 BOUNDS ACROSS STRIKES

The bounds so far constrain one option. Three further relations constrain options at *different* strikes on the same underlying and maturity, and they matter disproportionately: they are the conditions a whole option chain must satisfy, and Chapter 19 will use them as the test of whether a volatility surface is arbitrage-free. Write $c(K)$ for the price of a call struck at K, with $K_1 < K_2 < K_3$.

Monotonicity. $c(K_1) \ge c(K_2)$. A call with a lower strike confers the right to buy more cheaply, so it cannot be worth less. If it were, buy the low-strike call, sell the high-strike one, collect the difference, and hold a spread whose payoff is never negative.

The spread bound. $c(K_1) - c(K_2) \le (K_2 - K_1)e^{-rT}$. The payoff difference between the two calls never exceeds $K_2 - K_1$, so their price difference cannot exceed its present value.

Convexity. For equally spaced strikes, $K_2 = (K_1 + K_3)/2$,

$$c(K_1) + c(K_3) \ge 2\,c(K_2) \tag{11.3}$$

Buying the outer two calls and selling two of the middle one is a **butterfly**, and its payoff at expiration is zero outside $[K_1, K_3]$ and positive inside, peaking at K_2. A position whose payoff is never negative cannot be established for a credit, which is exactly what (11.3) says. With monotonicity, this makes $c(K)$ a convex, decreasing function of K, the shape drawn in Figure 11.3. Monotonicity and convexity hold for American calls too, since the long calls can be exercised whenever a short one is assigned; the spread bound loses its discount, $C(K_1) - C(K_2) \le K_2 - K_1$, because assignment can demand $K_2 - K_1$ at once.

Geometrically, convexity says the curve lies below any chord joining two of its points. The vertical gap between the chord and the curve at K_2 is half the value of the butterfly, since the butterfly sells the middle call twice, and it must be nonnegative.

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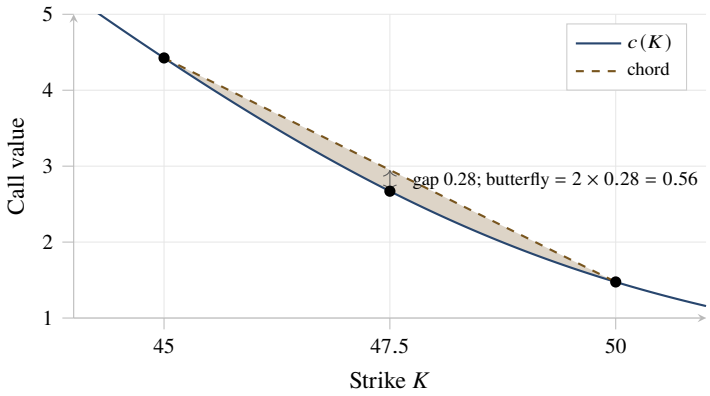


Figure 11.3: Convexity in the strike, at the three strikes of Example 11.2. Shaded: the gap between chord and curve.

Example 11.2 Testing a chain for convexity

Take the calls from Table 10.2, using mid prices. Options on individual US stocks are American, and monotonicity and convexity hold for both styles, so write $C(45) = 4.425$, $C(47.50) = 2.67$, $C(50) = 1.475$. The strikes are equally spaced, so (11.3) requires

$$4.425 + 1.475 = 5.90 \;\geq\; 2 \times 2.67 = 5.34$$

The chain passes, with \$0.56 to spare, which is the butterfly's value.

Check the other two conditions as well. Monotonicity holds: 4.425 > 2.67 > 1.475. Because the options are American, the spread bound takes its undiscounted form:

$$C(45) - C(50) = 2.95 \;\leq\; 50 - 45 = \$5$$

with about \$2 to spare.

Now suppose the 47.50 call were instead quoted 3.00 bid / 3.10 ask. Convexity would fail: $2 \times 3.05 = 6.10$ against 5.90. Trade it at the quotes: buy the 45 call at its \$4.50 ask, buy the 50 call at its \$1.51 ask, sell two 47.50 calls at the \$3.00 bid:

$$\text{net cash} = 2 \times 3.00 - 4.50 - 1.51 = 6.00 - 6.01 = -\$0.01$$

That is a penny of cost, not a credit, so the violation at mid prices does *not* survive the bid–ask spread, and there is no trade. Had the 47.50 bid been \$3.05, the position would have been established for a \$0.09 credit with a guaranteed payoff between \$0 and \$2.50: an arbitrage.

11.4 PUT–CALL PARITY

The chapter's central result relates the call and the put at the same strike and maturity, and it compares Portfolios A and C of Section 11.2. At expiration, A is worth $\max(S_T, K)$ as shown there. Portfolio C is worth S_T if $S_T > K$ (the put lapses) and K if $S_T \leq K$ (the put is exercised), also $\max(S_T, K)$. Two portfolios worth the same at T in every state must cost the same today:

$$\boxed{c + Ke^{-rT} = p + S_0} \tag{11.4}$$

This is **put–call parity**. It holds for European options on a non-dividend-paying stock, and it requires no assumption whatever about how the stock behaves. If dividends with present value D are expected during the option's life, the share in Portfolio C also delivers those, and the relation becomes

$$c + D + Ke^{-rT} = p + S_0 \tag{11.5}$$

Because $c - p$ is linear in K in either form, a put butterfly is worth what the call butterfly at

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the same strikes is worth, so puts are convex in the strike too, and the call's monotonicity and spread bound become $0 \le p(K_2) - p(K_1) \le (K_2 - K_1)e^{-rT}$. Example 11.3 shows what a violation of (11.4) is worth and how it is traded.

Example 11.3 Trading a parity violation

A stock trades at \$31 and pays no dividends. Three-month European options struck at \$30 are quoted: the call at \$3.00, the put at \$2.25. The rate is 10%.

Parity (11.4) says the put should be

$$p = c + Ke^{-rT} - S_0 = 3.00 + 30e^{-0.025} - 31 = \$1.2593$$

The put is quoted \$0.99 too high. Sell it, and hedge by buying the replicating portfolio: buy the call and short the stock, investing the proceeds.

$$\text{cash in} = 2.25 - 3.00 + 31 = \$30.25, \qquad 30.25\, e^{0.025} = \$31.0158$$

At expiration, if $S_T > 30$, exercise the call. If $S_T \le 30$, the put is exercised against you and you buy at \$30. Either way the short is closed for \$30, leaving

$$31.0158 - 30 = \$1.0158$$

in every state. Again this is the mispricing carried forward: $0.9907 \times e^{0.025} = \$1.0158$.

This trade has a name. Long call, short put, short stock is a **reversal**; the opposite is a **conversion**.

Parity for other underlyings

Chapter 10 cataloged options on indices, currencies, and futures. Parity extends to each of them by replacing the share in Portfolio C with whatever position replicates the underlying at time T, and the results are worth having explicitly:

$$\text{Index, dividend yield } q: \quad c + Ke^{-rT} = p + S_0 e^{-qT} \tag{11.6}$$

$$\text{Currency, foreign rate } r_f: \quad c + Ke^{-rT} = p + S_0 e^{-r_f T} \tag{11.7}$$

$$\text{Futures, price } F_0: \quad c + Ke^{-rT} = p + F_0 e^{-rT} \tag{11.8}$$

Equations (11.6) and (11.7) are the same statement: an index yielding q and a currency yielding r_f both require holding less than one unit today to have one unit at T, which is the reasoning of Sections 5.4 and 5.7 carried into option pricing. The futures version (11.8) rearranges to something that holds for every underlying once F_0 is read as its forward price to T, and is used constantly on trading desks:

$$c - p = (F_0 - K)e^{-rT} \tag{11.9}$$

Set $K = F_0$ and the right-hand side vanishes: a call and a put struck at the forward price have equal value. This is why the strike equal to the forward price, called **at-the-money forward**, is the usual reference point for quoting options, and why a straddle at that strike (Chapter 12) is the standard way to trade volatility without taking a directional position: the call and the put cost the same, and their deltas (Chapter 18) nearly offset. Chapter 19 measures moneyness and log-moneyness from this strike.

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11.5 SYNTHETIC POSITIONS

Equation (11.4) has four terms, so it can be rearranged four ways, and each rearrangement is a recipe for manufacturing one instrument from the others:

$$S_0 = c - p + Ke^{-rT} \qquad \text{synthetic long stock} \tag{11.10}$$

$$c = p + S_0 - Ke^{-rT} \qquad \text{synthetic call}$$

$$p = c - S_0 + Ke^{-rT} \qquad \text{synthetic put}$$

$$-S_0 = p - c - Ke^{-rT} \qquad \text{synthetic short stock} \tag{11.11}$$

Figure 11.4 shows the first of these without its cash: a long call combined with a short put at the same strike has exactly the payoff of a long forward at K. Equation (11.11) is the synthetic short Section 10.7 promised, whose put trades above parity by the borrow fee (Example 10.4).

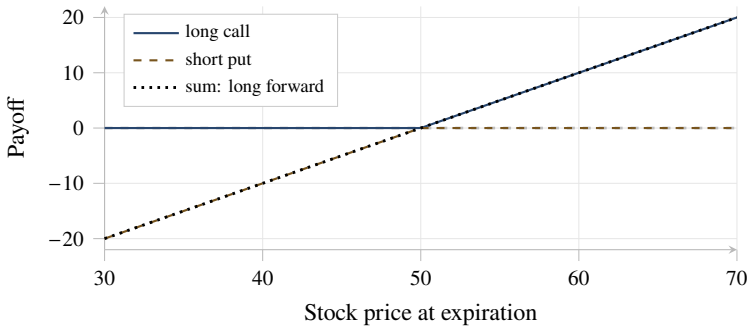


Figure 11.4: Long call plus short put at the same strike. $K = 50$.

11.6 READING THE IMPLIED FORWARD

Parity is usually taught as a test. In practice it is a *measurement device*: rearranged, it extracts from observable option prices a quantity no one quotes directly and everyone needs.

Rearranging (11.9) gives the option market's own forward price,

$$F_0 = K + (c - p)\, e^{rT} \tag{11.12}$$

It requires no view on dividends, no borrow-cost assumption, and no model.

Example 11.4 What the chain says about dividends

Take the 47.50 strike from Table 10.2: call mid 2.67, put mid 2.575, 45 days to expiration, $r = 5\%$. From (11.12),

$$F_0 = 47.50 + (2.67 - 2.575)\, e^{0.05\times 45/365} = \$47.596$$

The spot price is \$48.30, so the option market's 45-day forward is 70 cents below spot. That is striking, because interest alone would put the forward *above* spot, by $48.30\,(e^{0.05\times 45/365} - 1) = \$0.30$. Something is pulling it down by a dollar.

That something is mostly dividends, the roughly \$1.00 that Example 10.3 found from the same quotes (1.092 − 0.095). So the chain says the market expects about \$1.00 of dividends before expiration, roughly 2% of the share price in 45 days, which suggests a high-yielding stock with an ex-dividend date inside the option's life. The option prices contain the market's forecast, and where that forecast differs from the announced dividend, the gap is itself information: about a dividend the market expects to change, or, as the second caveat explains, about the cost of borrowing the stock.

Two caveats keep this honest. First, the options of Table 10.2 are American, and Section 11.11

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shows that arbitrage alone confines $C - P$ here only to a band about \$1.29 wide. The reading is usable because at 47.50, nearest the money, both options are liquid and neither carries much early-exercise premium, so $C - P$ stays close to the European $c - p$; the \$1.00 is good to ten cents or so, not to the cent. Second, parity gives *one* equation, and the forward embeds dividends, borrow costs, and financing together. Having computed \$1.00, one cannot separate "dividends of \$1.00" from "dividends of \$0.60 plus a 40-cent borrow cost" without outside information. Practitioners therefore quote the implied forward and stop there: decomposing it further is a modeling choice, not a measurement.

11.7 BOX SPREADS: PARITY AS A FINANCING TRADE

Combining a synthetic long at one strike with a synthetic short at another produces something with no equity exposure at all. Buy the call and sell the put at K_1; sell the call and buy the put at K_2, with $K_1 < K_2$. From (11.10) and (11.11) the stock cancels, and the payoff at expiration is $K_2 - K_1$ regardless of where the stock finishes. This is a **box spread**, and it is a zero-coupon bond assembled from four options; Example 11.5 reads off the interest rate its price implies.

Example 11.5 Borrowing and lending through the option market

An investor constructs a box on a stock index using one-year European options at strikes 4,900 and 5,000. Whatever the index does, the position pays 100 index points at expiration, \$10,000 per contract at the \$100 multiplier.

If the box can be bought for 95.12 index points (\$9,512 per contract), the implied lending rate is

$$r = \ln\left(\frac{100}{95.12}\right) = 5.00\%$$

If it can be bought for 93.00 points, the rate is $\ln(100/93.00) = 7.26\%$, and the investor lends at 7.26% on a fully collateralized, centrally cleared position. *Selling* the box borrows at that rate.

Institutions use box spreads as a financing instrument, competing with repo, because the position is cleared, requires no counterparty credit assessment, and can be executed in large amounts on an exchange. The implied rate is a directly observable financing rate, worth comparing against the rates of Chapter 4, since discrepancies between them are the option market's version of the basis trades of Chapter 6.

Case Study 11.1 The box that was not riskless

A box spread is riskless *if the options are European*. In 2019 a retail trader on a US brokerage constructed box spreads using options on an ETF (which, as Section 10.3 noted, are American) and treated the position as a guaranteed loan. The account was reported to have lost more than ten times its \$5,000 of capital.

The mechanism is exactly the assignment risk of Section 10.9. Two of the four legs are short options. If the ETF moves so that one of the short options is deep in the money, its holder may exercise it early. The trader is then assigned, must deliver or receive shares immediately, and is left holding a position that no longer has the offsetting structure the box depended on—at a moment chosen by someone else, not by them.

Two lessons follow, and both generalize past this episode. First, every result so far must be re-examined before it is applied to American options. Second, a position that is riskless at expiration is not necessarily riskless *before* expiration. Chapter 3 made this point about funding and Chapter 6 about the cash–futures basis trade. Here it appears as a counterparty's right to exercise early, when it suits them rather than you.

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11.8 EARLY EXERCISE: AMERICAN CALLS WITHOUT DIVIDENDS

An American call on a non-dividend-paying stock should never be exercised early. The result is stronger than it first appears, and it has two independent proofs worth knowing because they generalize differently.

The bounds argument. From (11.1), an American call is worth at least $S_0 - Ke^{-rT}$, since it is worth at least as much as the European call. Exercising yields only $S_0 - K$. Because $Ke^{-rT} < K$ whenever $r > 0$,

$$\underbrace{S_0 - Ke^{-rT}}_{\text{least value if sold}} > \underbrace{S_0 - K}_{\text{value if exercised}}$$

The holder always does better selling the option than exercising it. Note what this proves: at every date before expiration the call is worth more alive than exercised, so the right to exercise early is never used and adds nothing, and $C = c$ for a non-dividend-paying stock.

The economic argument. Exercising early does two things, both bad. It pays K now rather than at T, giving up the interest, $K - Ke^{-rT} = K\left(1 - e^{-rT}\right)$, which is the difference between the two sides of the display above. And it discards insurance: a holder who exercises owns the stock outright and suffers fully if it falls, whereas a holder who waits keeps a floor at zero. The second point is why the result holds even at $r = 0$, where the interest argument vanishes. Example 11.6 puts numbers on both.

Example 11.6 Why not to exercise

A stock trades at \$50 and pays no dividend. An investor holds an American call struck at \$40 with one month to run. The rate is 6%. Exercising immediately yields $50 - 40 = \$10$.

But the call must be worth at least

$$50 - 40e^{-0.06\times 1/12} = 50 - 39.800 = \$10.20$$

so selling it yields at least \$10.20: twenty cents more, guaranteed, with no view on the stock.

Suppose instead the investor wants to own the stock and holds the call for that reason. Even then, exercising is wrong: they should sell the call for at least \$10.20 and buy the stock at \$50, which costs \$39.80 net against \$40 for exercising. And if what they want is to keep the exposure for the remaining month rather than own the stock, holding the call is better still, because exercising would also throw away the protection against the stock falling below \$40 during that month.

In a frictionless market no preference makes exercising right, which is unusual: most decisions in finance depend on what the investor wants.

11.9 EARLY EXERCISE: AMERICAN PUTS

For puts the conclusion reverses, and seeing why shows what the call case relied on. Take the extreme case. A stock has fallen to \$0.05 and an American put struck at \$50 has six months to run. Exercising now yields \$49.95 immediately. Since the stock cannot fall below zero, waiting can add at most the last five cents, and it costs interest on the strike for as long as the holder waits: held to expiration, the \$50 is worth $50e^{-rT}$ today, less than \$49.95 at any rate above about 0.2%. Waiting a shorter time pays only if the stock reaches zero almost at once, which no realistic model allows. Exercise now.

The general principle is that exercising a put brings forward the receipt of K, which earns interest, whereas exercising a call brings forward the payment of K, which costs interest. The insurance argument still cuts the other way (exercising discards the chance of the stock falling further), but for a sufficiently deep in-the-money put the interest gain dominates.

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There is therefore a **critical stock price** $S^*(t)$ below which immediate exercise is optimal, and it rises toward K as expiration approaches. Figure 11.5 shows the consequence for value.

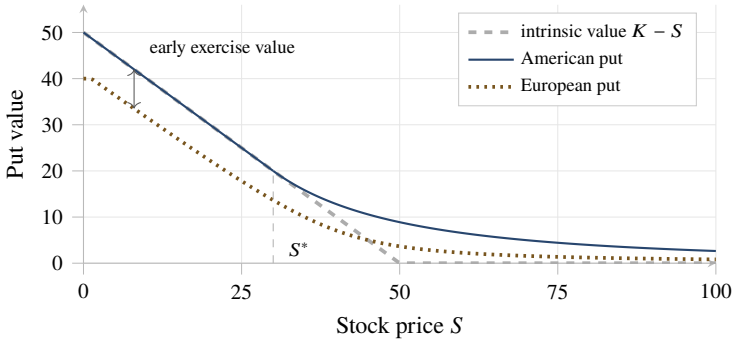


Figure 11.5: Put values against the stock price. $K = 50$; schematic, with the gap between the two exaggerated.

Three features of Figure 11.5 carry the argument. The American put coincides with intrinsic value below S^*, where it is exercised. The European put falls *below* intrinsic value for low stock prices (it is worth less than $K - S$ because the K arrives only at T), which is possible precisely because it cannot be exercised. And the European put is capped at Ke^{-rT} while the American put is capped at K. The gap between the two curves is the value of the right to exercise early, and unlike the call case it is genuinely positive: $P > p$.

11.10 DIVIDENDS AND EARLY EXERCISE

Now consider the third question Chapter 10 deferred. Ordinary cash dividends do not adjust the contract (Section 10.2), so an option holder simply suffers the price drop on the ex-dividend date. In a frictionless market that, and only that, can make early exercise of an American call rational. Frictions supply a second reason: a stock that is costly to borrow earns its owner a lending fee that a call holder does not receive, so it behaves like a stock paying a dividend, and a deep in-the-money call on it can be worth exercising early.

The trade-off is sharp. Exercising just before the stock goes ex-dividend captures the dividend D, because the exerciser owns the stock across the ex-date. The costs are the downside protection given up and the interest forgone on K over the remaining life $T - t$, where t is just before the ex-dividend date (only up to the next ex-dividend date, if one falls before expiration). Comparing the dividend with the interest alone gives a clean condition: early exercise cannot be optimal unless

$$D > K\left(1 - e^{-r(T-t)}\right) \tag{11.13}$$

Example 11.7 applies the condition to a holder facing an ex-dividend date.

Example 11.7 Should the call be exercised for the dividend?

A stock trades at \$60 and goes ex-dividend tomorrow, paying \$1.50. An investor holds American calls struck at \$55 with two months remaining. The rate is 5%.

The interest forgone by paying the strike two months early is

$$K\left(1 - e^{-r(T-t)}\right) = 55\left(1 - e^{-0.05\times 2/12}\right) = \$0.456$$

The dividend is \$1.50, comfortably larger. Condition (11.13) is satisfied and exercise may well be optimal: capturing \$1.50 outweighs giving up \$0.46 of interest.

Now change one number. If the dividend were \$0.30 instead, the condition fails (\$0.30 against

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\$0.456) and exercise is *definitely* wrong, whatever else is true.

Condition (11.13) is necessary, not sufficient. When it holds, exercise *may* be optimal, but the comparison above ignored the insurance the holder abandons, which is worth something whenever the stock might fall below \$55 in the remaining two months. A holder deep in the money, where that protection is nearly worthless, should exercise; a holder near the money should think harder.

The practical rule is that early exercise of American calls, when it happens at all, happens immediately before an ex-dividend date and, borrow fees aside, essentially never otherwise, which is why option desks maintain dividend calendars.

For American puts, dividends work the opposite way. A dividend lowers the stock price, which raises the put's value and makes waiting more attractive. Dividends therefore *discourage* early exercise of puts, and the critical price S^* falls as a dividend approaches.

11.11 PARITY BOUNDS FOR AMERICAN OPTIONS

Put–call parity is an equality for European options and fails for American ones, because either side may be exercised at a time the arbitrage argument does not control. What survives is a pair of inequalities. For a non-dividend-paying stock,

$$S_0 - K \;\le\; C - P \;\le\; S_0 - Ke^{-rT} \tag{11.14}$$

For the right-hand inequality, hold Portfolio C and be short Portfolio A, both in American form: hold the put and one share, write the call, and borrow Ke^{-rT}. If the call is assigned early, the share is delivered for K, which more than repays the loan, and the put is kept; otherwise everything nets to zero at T, and any dividends collected along the way are extra. So $P + S_0 - C - Ke^{-rT} \ge 0$, with or without dividends. The left-hand inequality follows from a portfolio argument of the same kind. With dividends of present value D, only the left edge moves, down by D:

$$S_0 - D - K \;\le\; C - P \;\le\; S_0 - Ke^{-rT} \tag{11.15}$$

The practical content is that $C - P$ sits in a band of width $K - Ke^{-rT}$, the interest on the strike over the option's life, plus D when the stock pays a dividend. For the Table 10.2 stock at the 47.50 strike, with $D = \$1.00$, the band runs from $48.30 - 1.00 - 47.50 = -\$0.20$ to $48.30 - 47.208 = \$1.09$, \$1.29 wide; the European value $S_0 - D - Ke^{-rT}$ is \$0.092, which the quoted \$0.095 matches by construction, since D was read from that quote; arbitrage alone would have allowed an American $C - P$ anywhere from \$0.29 below it to \$1.00 above. Table 11.2 collects every bound in the chapter: the European difference is a point, and the American one a band around it.

Table 11.2: Bounds from arbitrage alone; D is the present value of dividends (0 if none).

	Lower bound	**Upper bound**
European call c	$\max(S_0 - D - Ke^{-rT},\ 0)$	S_0
American call C	$\max(S_0 - D - Ke^{-rT},\ S_0 - K,\ 0)$	S_0
European put p	$\max(D + Ke^{-rT} - S_0,\ 0)$	Ke^{-rT}
American put P	$\max(D + Ke^{-rT} - S_0,\ K - S_0,\ 0)$	K
$c - p$	$S_0 - D - Ke^{-rT}$	$S_0 - D - Ke^{-rT}$
$C - P$	$S_0 - D - K$	$S_0 - Ke^{-rT}$

11.12 WHAT VIOLATIONS ACTUALLY MEAN

Everything above assumed frictionless trading. Real chains appear to violate these relations routinely, and reading those violations correctly is the practical skill this chapter is for.

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Borrow costs. A borrow fee opens a band above parity (Example 10.4). Inside it there is no arbitrage, and puts priced above parity are quoting the borrow fee, not a bearish view.

Transaction costs. A conversion involves four legs (call, put, stock, financing), each crossing a spread. Table 10.2's spreads at the 50 strike, \$0.07 on the call and \$0.15 on the put, cost half of each, about \$0.11, against mid prices on the option legs alone, and twice that if the position is unwound early: more than many apparent violations.

The financing rate is not r. Every relation in this chapter used the risk-free rate. Section 5.15 argued at length that no participant borrows at it, and the relevant rate is the one *this* participant can actually finance at. Two firms observing the same chain can correctly disagree about whether an arbitrage exists.

Dividends are estimates. Parity with dividends, (11.5), requires knowing D. For an option spanning an uncertain dividend, an apparent violation reflects a different dividend forecast rather than a mispricing.

The lesson is the one that recurs from Chapter 5: *these relations define a band, not a point*, and the width of the band is a real economic quantity. A trader who assumes it is zero and trades every apparent violation will discover the width empirically, which is more expensive.

11.13 WHO TRADES THESE RELATIONSHIPS

Several kinds of participant trade these relationships directly, and who they are explains why real chains are as consistent as they are.

Option market makers are the largest users. A market maker who has accumulated an unwanted position uses conversions and reversals (Section 11.4) to transform it (turning a short put into a long stock position plus a short call) at a cost they can compute in advance. This is not speculation; it is inventory management, and it is why quotes across a chain move together rather than independently.

Dispersion and volatility desks (Chapter 22) use parity to strip the directional component out of a volatility position.

Financing desks trade box spreads (Section 11.7); when box rates diverge from repo rates, the reason is usually balance-sheet capacity rather than mispricing, as Section 5.15 argued.

Structured-product desks use synthetics (11.10)–(11.11) to manufacture exposures they cannot trade directly. Where shorting is restricted, prohibited, or prohibitively expensive, the synthetic short is the only route.

Index and dividend traders use the implied forward of Section 11.6 as a traded quantity in its own right. Dividend expectations extracted from option chains are traded directly in dividend futures and dividend swaps, which pay the dividends actually distributed over a stated period; there the number computed in Example 11.4 is the price being traded.

The common thread is that none of these participants is forecasting the stock. They are trading the relationships among instruments written on it, and their competition is what makes the relationships hold.

SUMMARY

Every result in this chapter follows from the absence of arbitrage (a strategy that costs nothing, cannot lose, and can gain), without any model of how the stock behaves. The book assumes throughout that no such opportunity exists, as a statement about prices rather than about people.

Six factors determine an option's value. Calls and puts both rise with volatility, because the payoff is truncated at zero and greater dispersion carries no symmetric penalty; higher rates raise calls and lower puts; and more time helps American options unambiguously but not necessarily European ones, which cannot act on the extra time. Arbitrage alone then bounds each price above and below, and across strikes it imposes monotonicity, the spread bound and

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convexity. Convexity says a butterfly cannot have negative value, a test that must be run at prices one can trade rather than at mid prices, and Chapter 19 uses it to test a volatility surface for arbitrage.

Put–call parity, $c + Ke^{-rT} = p + S_0$, links the call and put at the same strike and maturity. Rearranged, it manufactures synthetic stock and synthetic options, and the synthetic short explains why puts trade above parity on hard-to-borrow stocks; two synthetics at different strikes make a box spread, a zero-coupon bond built from options that is used as a financing instrument and is riskless only when the options are European. Parity is also a measurement: the implied forward read off an option chain is the chain's most useful single number, and its shortfall from the carried-forward spot reflects mostly expected dividends, mixed with borrow costs.

An American call on a non-dividend-paying stock should never be exercised early, because selling always beats exercising: the option is worth more than its exercise value, and exercising discards downside protection. An American put may be exercised early, because deep in the money the interest earned on receiving the strike now outweighs the option value given up, so below a critical stock price exercise is optimal. In a frictionless market, dividends are the only reason to exercise an American call early, and when they make it optimal, it happens immediately before an ex-dividend date.

For American options parity becomes a pair of inequalities rather than an identity. In practice all these relations define bands rather than points, widened by borrow costs, transaction costs, financing that is not at the risk-free rate, and uncertain dividends. The width is an economic quantity worth measuring, not an error term to ignore.

FURTHER READING

Merton's 1973 paper on rational option pricing established most of the results in this chapter and states them in full. Stoll's 1969 paper introduced put–call parity to the modern literature. Natenberg's *Option Volatility and Pricing* describes conversions, reversals, and boxes as a market maker trades them. Ofek, Richardson and Whitelaw (2004) is the evidence for Section 11.12: apparent parity violations track borrow costs and short-sale constraints. Jensen and Pedersen (2016) is the place to start on when frictions make early exercise of a call rational, and van Binsbergen, Diamond and Grotteria (2022) read risk-free rates off index-option prices through the parity relation a box spread locks in.

PRACTICE QUESTIONS

11.1 Explain why both a call and a put increase in value with volatility, and why a forward contract's value does not depend on volatility at all.

11.2 A non-dividend-paying stock trades at \$45. A six-month European call struck at \$42 trades at \$3.10 and the rate is 6%. Is the lower bound violated? If so, construct the arbitrage and compute the profit.

11.3 A stock trades at \$28 with no dividends. Four-month European options struck at \$27 are quoted: call \$2.60, put \$1.10, rate 5%. Test parity and, if it fails, state the trade and the profit.

11.4 Using put–call parity, express a synthetic long put in terms of a call, the stock, and cash. Verify the payoff in both states at expiration.

11.5 Explain why an American call on a non-dividend-paying stock should never be exercised early, giving both the bounds argument and the economic argument. Which of the two still works when $r = 0$?

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11.6 A stock has fallen to \$0.25 and an American put struck at \$40 has three months remaining. Should it be exercised? Give the numerical comparison with $r = 4\%$.

11.7 A stock at \$80 goes ex-dividend tomorrow, paying \$0.90. An investor holds American calls struck at \$75 with three months remaining; the rate is 4%. Apply condition (11.13) and state what conclusion is and is not justified.

11.8 Construct a box spread from options struck at \$40 and \$50 expiring in six months, list the four legs, and state the payoff. If the box costs \$9.70, what rate is implied?

11.9 State the three relations that constrain call prices at different strikes on the same underlying and maturity, and for each identify the position that profits if it is violated.

11.10 Using the mid prices of Table 10.2 at every strike, with $r = 5\%$, compute the implied forward (11.12) and the dividend it implies, $S_0 - F_0 e^{-rT}$. At which strikes does the reading drift, in which direction, and why? Which strike would you quote?

11.11 State the parity bounds for American options, explain why the European equality becomes a band, identify which side of the band binds when the put is deep in the money and why, and state how the band changes when the stock pays dividends of present value D.

11.12 A chain shows puts trading 40 cents above parity on a \$50 stock with six months to expiration. Using Example 10.4, state the borrow fee that would account for it, and explain why observing a violation of this size is not the same as being able to profit from it.

FURTHER QUESTIONS

11.13 Section 11.1 states that more time to expiration can *reduce* the value of a European option. Construct two explicit examples, one for a call and one for a put, in which this occurs, and identify precisely which feature of European exercise makes it possible.

11.14 The case study in Section 11.7 describes a box spread that was not riskless because it used American options. Work out exactly which of the four legs can be assigned early, under what stock price conditions, and what position the trader is left holding. Then state a rule for when a box may safely be treated as a financing trade.

11.15 Condition (11.13) is necessary but not sufficient. Construct a case in which the condition holds and early exercise is nevertheless wrong, and explain what quantity the condition omits. How would you decide the question in practice?

11.16 An option chain on a hard-to-borrow stock shows puts persistently above parity by an amount implying a 15% borrow fee. A colleague proposes selling puts and buying calls to "harvest the mispricing," delta-hedging with stock in the manner of Chapter 18. Explain what exposure this position actually carries, and what would have to happen for it to lose money.

11.17 Derive the left-hand inequality in (11.14) by constructing two portfolios and comparing their values. State which step uses the put's right to early exercise, and show that for European options the inequality follows from the equality (11.4).

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